Handover eddy-viscosity ratio $\chi = \nu_t/\nu$ at transition, from the XFOIL/Drela closure.

1. Equilibrium shear stress. XFOIL's equilibrium shear-stress coefficient, $C_\tau \equiv \tau_{t,\max}/\rho u_e^2$, is

$$C_{\tau,\mathrm{eq}} = \frac{CC\, H^*\,(H_k-1)\,H_{kc}^2}{(1-U_s)\,H\,H_k^2}, \qquad H_{kc} = H_k - 1 - \frac{G_C}{\mathrm{Re}_\theta} \xrightarrow[\mathrm{Re}_\theta\ \text{large}]{} H_k - 1,$$

with slip velocity and prefactor

$$U_s = \tfrac{1}{2}H^*\Big(1 - \tfrac{1}{G_B}\tfrac{H_k-1}{H}\Big), \qquad CC = \frac{1}{2G_A^2 G_B} = \frac{1}{2(6.7)^2(0.75)} = 0.01485,$$

and $G_A = 6.7$, $G_B = 0.75$, $G_C = 18$; H^* is the turbulent KE shape parameter and $H = H_k$ incompressibly. The familiar 0.015 is CC.

2. The layer does not start at equilibrium. At the transition station XFOIL initializes

$$C_\tau(s_t) = c^2\, C_{\tau,\mathrm{eq}}, \qquad c = C_{\tau C}\, e^{-C_{\tau E}/(H_k-1)}, \qquad C_{\tau C} = 1.8, \quad C_{\tau E} = 3.3.$$

$c < 1$ encodes that high-amplitude TS waves carry only part of the eventual Reynolds stress: $c = 0.0024$ at $H_k = 1.5$, rising to $c = 0.48$ at $H_k = 3.5$.

3. Mapping C_τ to an eddy viscosity. Equating the definition of C_τ to Boussinesq and writing the peak shear in wall-free units,

$$\rho u_e^2 C_\tau = \rho\,\nu_{t,\max}\Big(\frac{\partial u}{\partial y}\Big)_{\max}, \qquad \Big(\frac{\partial u}{\partial y}\Big)_{\max} \equiv G\,\frac{u_e}{\theta}, \qquad \text{so} \qquad \boxed{\chi = \frac{\nu_{t,\max}}{\nu} = \frac{\mathrm{Re}_\theta\, C_\tau}{G}}$$

Justification of G (canonical profile). G is not assumed: it is measured on the *Falkner–Skan* family — the right canonical family here, since at a bubble's transition station the profile is still the laminar near-/post-separation shear layer. In similarity variables ($\eta = y/L$, $u = u_e f'$), $\theta = L\int f'(1-f')\,d\eta$, so $G = \big(\int f'(1-f')d\eta\big)\max f''$. Along the family:

H_k	2.68	2.80	2.96	3.18	3.48	3.98	4.92
G	0.208	0.203	0.203	0.205	0.209	0.211	0.207

$G = 0.207 \pm 2\%$ across the entire inflectional range — mild adverse gradient, through separation ($H_k = 3.98$), into reversed flow ($H_k = 4.92$). The peak shear is therefore *essentially independent of* H_k: as the profile deepens the shear layer thickens in proportion to its defect, and the two cancel in θ units. It does *not* scale as $(H_k-1)/H_k$, which would rise 25% here and (with $k_g \approx 0.6$) overstate the shear twofold. Hence $\chi \approx 4.8\,\mathrm{Re}_\theta\, C_\tau$.

4. Equilibrium versus handover level. The boxed mapping of step 3, evaluated at the two values of C_τ already established — the step-1 equilibrium and the step-2 initialization — gives

$$\chi_{\mathrm{eq}} = \frac{\mathrm{Re}_\theta\, C_{\tau,\mathrm{eq}}}{G}, \qquad \chi_{\mathrm{init}} = \frac{\mathrm{Re}_\theta\, c^2 C_{\tau,\mathrm{eq}}}{G} = c^2\chi_{\mathrm{eq}}.$$

Only χ_{init} answers "how much χ at handover": χ_{eq} is merely where the lag equation is heading.

regime	H_k	Re_θ	c	χ_{eq}	χ_{init}
flat plate (turb)†	1.5	500	0.00245	3.9	2.35×10^{-5}
flat plate (turb)†	1.5	1000	0.00245	8.3	4.97×10^{-5}
bubble (mild)	2.5	200	0.199	5.1	0.204
bubble (mild)	2.5	800	0.199	22.0	0.874
bubble (strong)	3.5	200	0.481	7.7	1.78
bubble (strong)	3.5	800	0.481	32.1	7.41

†Outside the Falkner–Skan calibration: an attached turbulent profile peaks nearer the wall where shear is larger, so these are upper bounds.

5. Consequence for the handover. The factor c carries *no* N dependence: it is a function of H_k alone, applied once at the station where N first reaches N_{crit}, and $c = 1$ only at $H_k = 1 + 3.3/\ln 1.8 = 6.6$. So XFOIL declares transition with the layer holding just c^2 of its equilibrium stress — 4% at $H_k = 2.5$, 23% at $H_k = 3.5$ — giving a handover level $\chi_\star = \chi_{\mathrm{init}} = O(0.2\text{–}7)$. It rises steeply with both bubble strength and Re_θ (about eightfold from $H_k = 2.5$ to 3.5 at fixed Re_θ), so "bigger bubble $\Rightarrow$ later handover" is quantitative. Note the magnitude: XFOIL's own initialization hands over at χ of order a few, *not* ≈ 30 — the layer is expected to develop the rest by itself, which is what SA is supposed to do.

Reproduced by `repro/analytic/handover_viscosity_ratio.py`, cross-checked to 10^{-9} against `get_cteq/get_cttr` in `src/validation/mfoil.py`.